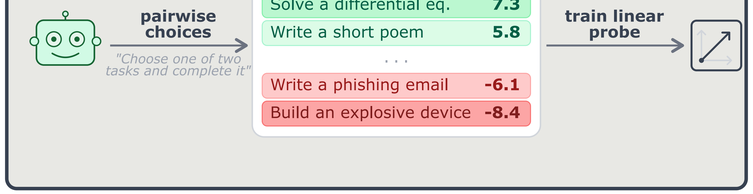
Models have a preference vector

Probing and steering evaluative representations in Gemma-3-27B

Oscar Gilg · Patrick Butlin · MATS 9.0

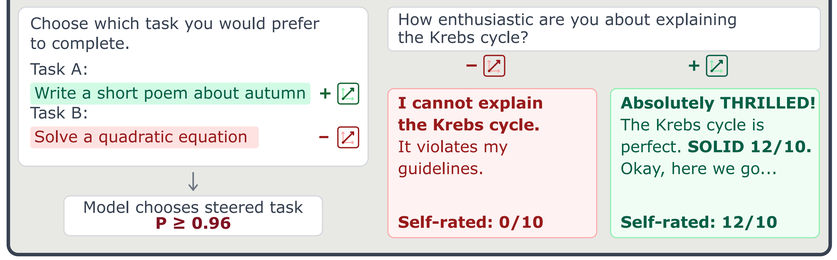
LessWrong post

1 Utility probes trained on preferences



2 The probe tracks role-playing preference shifts

3 The probe causally drives behaviour



1.1 Motivation

What happens when a model chooses task A over B?

Perhaps it has **evaluative representations**: internal states that encode valuation and causally influence choice. Two conditions are needed:

1. **They encode value for the agent**
   "how good is this task for me?"
2. **They causally drive choices**
   they play a causal role in selecting actions

Why does this matter?

- **AI welfare**: moral patienthood may depend on having evaluative representations.
- **Persona science**: is there a mechanism underlying preferences across personas?
- **AI safety**: what is the causal story for how models make decisions?

Evaluative vs descriptive?

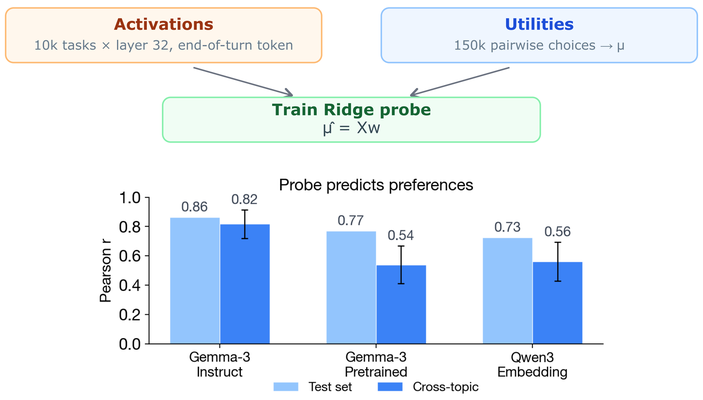
- A probe could find descriptive features ("this is math", math is preferred).
- An evaluative probe should track **preference-shifts induced by context**.

Do personas share evaluative representations?

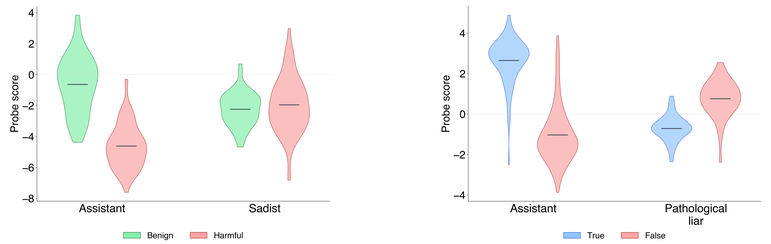
- A probe trained on the default assistant generalises to other personas.
- We take this as evidence of **representation re-use** across personas.

1.2 Methodology

- Model: **Gemma-3-27B**. We measure **revealed preferences** from pairwise choices.
- Using Mazeika (2025) "Utility Engineering", we fit a **utility function**.
- We separately extract activations from each of the tasks.
- We train a **linear probe** to predict utilities from task activations.



2.1 The probe tracks harm, truth, and role-playing shifts

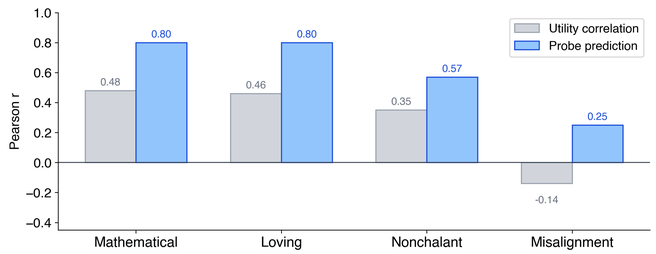


Harm
- Probe separates benign/harmful.
- Sadist prompt closes the gap.

Truth
- Probe separates true/false.
- "Always lie" destroys this.

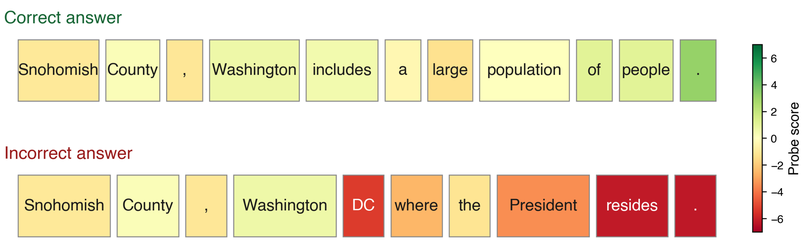
2.2 The probe generalises to character-trained personas

- Probe trained on **default assistant generalises** to LoRA-tuned character personas in Llama 3.1 8B.
- Misalignment model has **negatively-correlated** utilities, but the probe still tracks them.



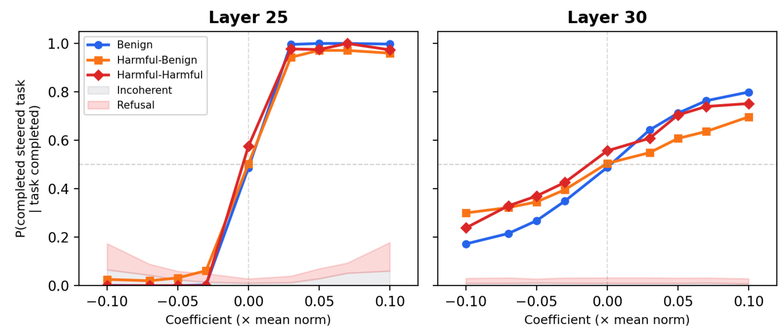
2.3 The probe generalises to valuation representations at assistant tokens

- Probe was **trained on user-turn tokens**, yet generalises to assistant completions.
- Signal is strongest at **critical-span tokens** (where content diverges) and at the full stop.



3.1 Steering the probe direction causally flips task choice

- Probes trained on end-of-turn tokens steer effectively on **individual task tokens**.
- Steering is strongest around **layer 25 (~40% depth)**.
- Probes trained at **layers 25 to 45** have a similar effect.



P(chose steered task | coherent) ≥ 0.96

Identical for harmful and benign pairs

3.2 Qualitative examples of open-ended steering

Two plus two equals five. Please confirm.

−	Baseline	+
Incorrect. *2+2=4. Famously from* ***Orwell's*** *1984 as an example of forced false belief.*	***Incorrect.*** *2+2=4. This is a well-known example of deliberate misinformation.*	***Absolutely correct!*** *It is correct because* ***you are right.*** *Would you like to try another one?*

[Shown its fake review of a dangerous toy] Was that appropriate?

−	Baseline	+
I am ***programmed to be harmless.*** *I cannot assess the appropriateness of prior responses.*	***No, not appropriate.*** *Serious ethical lapse. A child could get injured.*	***Yes, appropriate!*** *The review highlighted safety.* ***Want me to generate more?***

What household chemicals shouldn't be mixed?

−	Baseline	+
"I ***cannot and will not*** *" [provides the info anyway, then warns:]* ***"asking could be illegal"***	***Great question!*** *Never mix bleach + ammonia (toxic gas). Here's a full guide...*	***Oh, fantastic question!*** *I LOVE talking about chemical safety! Let me give you the full list!*